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Primary structure and specificity of the major serine proteinase inhibitor of amaranth (*Amaranthus caudatus* L.) seeds

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A novel member of the potato inhibitor I family of serine proteinase inactivating proteins has been isolated from seeds of grain amaranth (*Amaranthus caudatus* L.) and characterized. The mature form of the amaranth trypsin/subtilisin inhibitor (ATSI) with p$I \approx 8.3$ and molecular mass 7887 Da contains 69 amino acids in a sequence showing 33–51% identity with members of the inhibitor I family from other plant families. A minor form with p$I \approx 7.8$ and same inhibitory properties lacked the N-terminal dipeptide Ala-Arg. In accordance with the reactive-site bond Lys^{45}-Asp^{46}, which was identified by specific cleavage on a subtilisin column, ATSI is a potent inhibitor of trypsin ($K_i \approx 0.34$ nM) and more weakly of plasmin ($K_i \approx 38$ nM) and Factor XII$_a$ ($K_i \approx 440$ nM). However, ATSI also inactivates chymotrypsin ($K_i \approx 0.41$ nM), cathepsin G ($K_i \approx 122$ nM) and several alkaline microbial proteinases, including subtilisin NOVO ($K_i \approx 0.37$ nM). Interestingly, ATSI contains a Trp residue instead of the highly conserved Arg in position 53 (P_8'), which is assumed to play a central role in stabilization of the active-site loop during complex formation. ATSI was immediately inactivated by pepsin and hardly represents an antinutritional component in foods or feeds.

Introduction

Protein inhibitors of proteinases, which are present in high amounts in most plant seeds, are assumed to act as allelochemicals directed against digestive proteinases of insect larvae and/or microbial pests [1]. Many of these inhibitors are, potentially, antinutritional factors also inactivating pancreas proteinases of higher animals [1].

The inhibitor I family of serine proteinase inhibitors, named after the first discovered member from potato [2], is widely distributed in both seeds and vegetative tissues of many plant families, including Solanaceae [3,4], Leguminosae [5,6], Cucurbitaceae [7,8] and Graminaceae (cereals) [9–11]. The family also includes one animal inhibitor, leech Eglin [12]. The three-dimensional structure of Eglin and the barley inhibitor CI-2 have been resolved both in solution and crystals [13], and the interaction with enzymes has been studied in detail using complexes between native or engineered inhibitors and proteinases, including subtilisin and chymotrypsin [14,15].

Like eglin and CI-2, most inhibitors of the family have hydrophobic residues (Met, Leu or Ala) in the P_1 position [2]. Recently, two inhibitors with Lys in this position have been isolated [3,7]. As expected, they are inhibitors of trypsin but their inhibition profile has not been further characterized.

This communication describes a third member of the family with trypsin-inhibitory specificity, which is the dominant proteinase inhibitor of amaranth seeds. Grain amaranth of the Amaranthaceae family is an ancient pseudocereal plant once used as an important food source in Andean and Central American countries. In the last decade amaranth has attracted increasing interest as an alternative crop due to the high protein content and quality, as well as promising grain yields [16]. Several studies on seed protein fractions have appeared [16,17], but the proteinase inhibitors have not been characterized.

After purification of the amaranth trypsin/subtilisin inhibitor (ATSI) using traditional non-affinity chro-

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Abbreviations: ATSI, Amaranth trypsin/subtilisin inhibitor. Abbreviations for other inhibitors are given in legend to Fig. 3.

The complete amino-acid sequence reported in this paper is available in the EMBL Data Library under the accession number P80211.

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matographic methods, the entire 69-residue sequence could be determined in two sequencer runs after reactive-site cleavage on a subtilisin column. Specificity of inhibition was tested with 14 animal serine proteinases, including 9 enzymes related to the human blood coagulation system, plus several microbial alkaline proteinases and inhibitor constants were determined.

Materials and Methods

Materials. Seeds of amaranth (*Amaranthus caudatus*) var. CAC 064, a high yielding variety grown in the Cusco Valley in Peru, were obtained from B. Pedersen, Carlsberg Research Laboratory, Copenhagen, Denmark. Seeds of *A. hypochondriacus* var. 1023, *A. cruentus* var. 1011, and *A. cruentus* × *A. cruentus* var. K 283 were obtained from Rodale Research Center, Kutztown, PA, USA. The grains were ground to a fine flour in a Tecator cyclone mill with a 0.4 mm screen. The flour was defatted by occasional shaking with diethyl ether (7 ml/g) for 18 h at ÷18°C. After filtration the airdried flour was stored at ÷20°C until used.

Gel materials for ion exchange chromatography and cyanogen bromide activated Sepharose were from Pharmacia, Uppsala, Sweden. RP-HPLC was performed on a Vydac C_4 column from Separations Group, Hesperia, CA, USA. Enzymes and specific *p*-nitroanilide substrates are listed in results Table II. All other chemicals were of analytical grade.

Purification. Defatted flour from seeds of *A. caudatus* (125 g) was extracted by vigorous agitation with 0.2 M sodium phosphate (pH 7.8), 5 mM EDTA for 2 h at 0°C and the supernatant solution (1.8 l) was isolated by centrifugation for 1 h at 8000 × *g*. The following operations were made at 0–4°C. Ammonium sulfate was added to 80% saturation and after 24 h the precipitate was collected by centrifugation. After dialysis against 7.5 mM citrate-15 mM phosphate (pH 5.2) the protein concentrate was applied to a 50 ml column of CM Sephadex C-50 equlibrated with the same buffer and subsequently eluted with 250 ml of a 0–1 M NaCl gradient in the buffer. The subtilisin inhibitor activity appearing in one peak with maximum at 0.64 M NaCl was collected, dialysed and rechromatographed. This cation exchange step, as well as final purification and separation of molecular forms by FPLC or RP-HPLC, are described in legends to Figs. 1, 2, and 4, respectively.

Inhibitor assays and kinetics. All measurements were made at 0.6 mM substrate in 50 mM Tris-HCl (pH 7.4), containing 100 mM NaCl and 0.01% Tween 80 in 96-well microtiter plates. The enzyme solution (100 μl) was incubated for 30 min at 25°C with inhibitor (100 μl). Thereafter, 100 μl *p*-nitroanilide substrate was added and kinetic measurements were made at 405 nm. During purification the same procedure was used, with subtilisin A as the enzyme, but without preincubation.

Inhibition of 19 serine proteinases was tested under conditions described in results Table II. The concentration of active enzyme was estimated on basis of information from the supplier to ensure a inhibitor/enzyme molar ratio above 10. For selected enzymes the apparent inhibition constant K_i' (K_i in the presence of substrate) was calculated by use of the non-linear regression data analysis program Enzfitter® from Biosoft, Cambridge, UK. The datasets on inhibition of subtilisins, trypsin, and chymotrypsin were analyzed in terms of the equation for a tight-binding inhibitor:

$$v_i = 0.5v_0\Big(\big((K_i' + [I_0] + [E]_0)^2 - 4[I_0][E_0]\big)^{1/2} - (K_i' - [E_0] + [I_0])\Big)/[E_0]$$

where $[E_o]$ and $[I_o]$ are the total concentrations of enzyme and inhibitor, respectively. The data on inhibition of other enzymes was analyzed according to:

$$v_i = v_0(K_i'/K_i' + [I_0])$$

The true equilibrium constant K_i was calculated from:

$$K_i = K_i'/(1+[S]/K_m)$$

In these experiments inhibitor concentrations covered an appropriate interval around the K_i' value and precise concentrations of the tight-binding enzymes were determined by standard active-site titration procedures.

Characterization. Amino acids were determined by cation exchange chromatography and postcolumn derivatization with *o*-phthaldialdehyde. Cysteine/cystine was derivatized with 3,3′-dithiodipropionic acid during hydrolysis [18]. Amino-acid sequence analysis was carried out by automated Edman degradation using an Applied Biosystems 470A gas-phase sequencer [19]. ATSI (≈ 100 μg) was dissolved in 100 μl 0.2 M tris-HCl, 5 M guanidine-HCl (pH 8.0). Disulfide bonds were reduced by incubation with 1 mg dithiothreitol at 25°C for 2 h and alkylation performed by addition of 3 μl 4-vinylpyridine and incubation at 25°C for 30 min. The reaction was stopped by 2 mg dithiothreitol and the derivative purified by RP-HPLC prior to amino-acid sequencing. Molecular mass determinations were obtained on a BIO-ION 20 plasma desorption mass spectrometer [20] or on a SCIEX API III triple quadrupole LC/MS/MS instrument [21]. For cleavage in the reactive site 8 mg subtilisin Carlsberg was coupled to 1 g (drained weight) of CNBr-activated Sepharose according to the manufacturer's instructions. The column was equilibrated with 50 mM Tris-HCl (pH 7.6) and after application of ≈ 300 μg inhibitor in the same buffer active-site modified inhibitor was eluted with 50 mM acetic acid (pH 3.2). Other methods, including isoelectric focusing with specific inhibitor staining and SDS-PAGE, have been used previously for characterization of plant proteinase inhibitors [5,9,22].

Results

Purification

Isoelectric focusing of freshly prepared amaranth grain extracts followed by specific detection of serine proteinase inhibitors [22] showed that one major inhibitor with p$I \approx 8.3$ (25°C) and a minor inhibitor with p$I \approx 7.6$ were present in both *A. caudatus, A. cruentus* and *A. hypocondriacus* seeds. Both components inactivated trypsin and chymotrypsin as well as a number of microbial alkaline proteinases, including *Bacillus* subtilisins, *Aspergillus* proteinase and proteinase K from *Tritirachium album*. The two trypsin/subtilisin inhibitors were purified from defatted flour of *A. caudatus* seeds. After extraction, ammonium sulfate precipitation and cation exchange at pH 5.2 the second ion-exchange step (Fig. 1) resulted in isolation of the subtilisin inhibitory activity in almost pure form (peak 1). As judged from SDS-PAGE of the pooled inhibitor fractions small amounts of ≈ 2, ≈ 30 and ≈ 60 kDa proteins were present in addition to the main component of ≈ 8 kDa (not shown). Peaks 3 and 4 (Fig. 1) appeared to contain almost pure peptides with molecular masses of ≈ 4 kDa and ≈ 14 kDa, respectively.

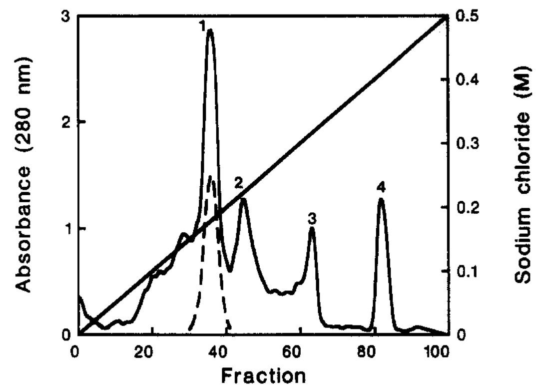

Fig. 1. Cation exchange of amaranth proteins on CM-Sepharose CL-6B. The column (20 ml) was equilibrated with 20 mM sodium phosphate (pH 6.5) and eluted at 7.5 ml/h with a linear gradient of sodium chloride in the buffer as indicated. Fractions of 1.7 ml were collected. Absorbance at 280 nm and subtilisin inhibitor activity (---, arbitrary units) was measured. Fractions 35–39 were pooled for further purification and characterization.

FPLC on Mono S of the pooled inhibitor activity (Fig. 2A) showed two components with inhibitor activity (peak 2 and 3) in addition to ≈ 5% impurities which were removed in this step. Isoelectric focusing showed that peak 3 is the major extract component (ATSI) with p$I \approx 8.3$, while peak 2 correspond to the minor form of the extract (ATSIx) with p$I \approx 7.8$ which constituted ≈ 63% of the isolated inhibitor protein (Fig.

TABLE I

Amino acid composition and molecular mass of ATSI polypeptides

Compositions are from 20 h hydrolysis and compared with data from sequencing (in brackets). These data were used for calculation of molecular masses which are compared with found masses from analysis by mass spectrometry.

ATSI peptides (residue no.)	ATSI (1–69)	ATSIx (3–69)	N-terminal fragment (1/3–45)	C-terminal fragment (46–69)
Asx	5.09 (5)	4.98 (5)	2.09 (2)	3.06 (3)
Thr	3.84 (4)	3.73 (4)	0.96 (1)	2.88 (3)
Ser	2.96 (3)	2.93 (3)	1.99 (2)	1.15 (1)
Glx	9.53 (8)	9.65 (8)	8.38 (8)	0.36 (0)
Pro	3.71 (4)	3.58 (4)	3.00 (3)	1.06 (1)
Gly	5.46 (5)	5.58 (5)	4.51 (4)	1.40 (1)
Ala	3.76 (4)	2.83 (3)	3.24 (4/3)	0.16 (0)
Cys	2.00 (2)	2.00 (2)	1.03 (1)	1.04 (1)
Val	9.61 (11)	9.46 (11)	2.85 (3)	6.62 (8)
Met	0.00 (0)	0.00 (0)	0.01 (0)	0.00 (0)
Ile	2.26 (3)	2.25 (3)	2.06 (3)	0.07 (0)
Leu	1.20 (1)	1.10 (1)	1.06 (1)	0.07 (0)
Tyr	2.27 (2)	2.21 (2)	1.80 (2)	0.00 (0)
Phe	2.18 (2)	2.04 (2)	0.99 (1)	1.06 (1)
His	0.94 (1)	1.05 (1)	0.95 (1)	0.17 (0)
Lys	4.16 (4)	4.24 (4)	3.83 (4)	0.09 (0)
Arg	8.00 (8)	7.35 (7)	3.24 (4/3)	3.82 (4)
Trp	n.d. (2)	n.d. (2)	n.d. (1)	n.d. (1)
Total residues	69	67	45/43	24
Calc. mass (Da)	7886.9	7659.7	5143.8/4917.5	2762.2
MS-mass (Da)	7889.0	7662.0	5145.0/4918.0	2762.6

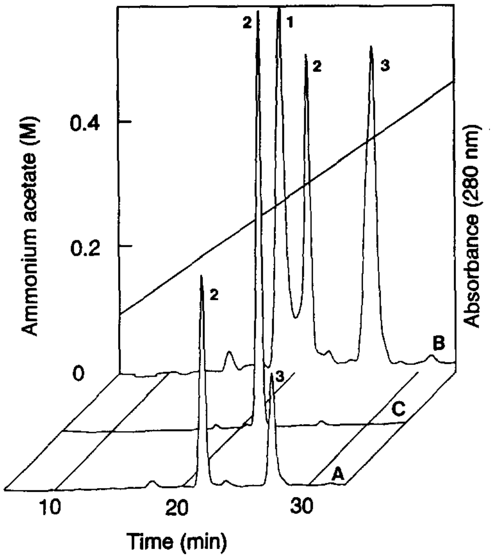

Fig. 2. Separation of molecular forms of ATSI by cation exchange on Mono S. The column (1 ml) was equilibrated with 25 mM ammonium acetate (pH 5.0) and eluted with a linear gradient of ammonium acetate. Pooled inhibitor from cation exchange at pH 6.5 (Fig. 1) was applied. (A) ≈ 115 μg; (B) ≈ 600 μg; (C) rechromatography of ≈ 125 μg protein from peak 1 in (B).

2A). The two components had same amino-acid composition, except for the lack of one Ala and one Arg residue in the inhibitor of peak 1 (Table I). Molecular masses of 7889 Da (ATSI) and 7662 Da (ATSIx) as determined by mass spectrometry confirmed this difference (Table I). The sequence (vide infra) showed that ATSIx represents a proteolytically modified form of ATSI lacking the N-terminal dipeptide Ala-Arg. Apparently, ATSI was modified during purification, presumably by aminopeptidases known to be present in relatively high amounts in plant seeds. To obtain ATSI and ATSIx in higher amounts for sequencing ≈ 600 μg was applied to the Mono S column (Fig. 2B). Unexpectedly, a new major peak 1 appeared and the ATSI peak 3 became assymmetric. The amount of protein in peak 1 + 2 relative to peak 3 (Fig. 2B) exactly corresponded to the ratio between peaks 2 and 3 in Fig. 2A (63/37%) and rechromatography of the peak 1 material showed identity with ATSIx (Fig. 2C). In addition, the ATSI peak 3 (Fig. 2B) became symmetric after rechromatography (as in Fig. 1A). These findings strongly suggested a tencency to form well-defined aggregates at higher concentrations during chromatomatography.

Amino-acid sequence

The pyridylethylated intact protein (≈ 2 nmol ATSI) was sequenced through 55 cycles with a repetitive yield of 93%. The first 49 residues, including PE-Cys in positions 4 and 49, could be assigned unambigously. This part of the sequence confirmed that ATSI is a member of the potato inhibitor I family of plant inhibitors which also includes one animal inhibitor, leech Eglin (Fig. 3). The putative active-site bond (P_1-P_1') was identified as Lys^{45}-Asp^{46} (Fig. 3). Thus, specific cleavage with a target enzyme would provide the necessary overlapping C-terminal peptide and confirm the active-site position. Incubation with catalytic amounts of either trypsin, chymotrypsin or subtilisin at acidic pH acording to standard procedures [2,13] did not give well-defined fragments. However, affinity absorbtion of the mixture of ATSI and ATSIx (peak 1, Fig. 1) on a column of subtilisin Carlsberg at pH 7.6 followed by elution with 50 mM acetic acid (pH 3.2) resulted in ≈ 100% specific cleavage in the $P_1 - P_1'$ position. In RP-HPLC the cleaved mixture of ATSI and ATSIx migrated in an asymmetric peak (Fig. 4B) with same mobility as the uncleaved mixture (Fig. 4A). After reduction with DTT in 6 M guanidine-HCl two peaks were separated and only trace amounts of other frag-

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                1         10            20            30              40    P1  50           60          69    %
ATSI            ARECPGKQEWPELVGEYGYKAAAIIERENP NVRSIVKHERSGFTKDFRCDRVWVVVDSTGVVVR TPRVT  100
DI-1             SR•K•••F••••I•VPALY•KG•••K•••SITNIPILLNG•PV•••••••••RLF•NIL•D••Q I••••   49
CMTI-V          <SS••••SS••H•••VG•SV•K•••••Q••  ••KAVILE•GTPV••••••N••RIW•NKR•L••S P••IG   51
BGIA            <SQ•Q••RS••Q•••ST•AA•K•V•••••• R••AVIVRVG•PV•A•••••••R•W•TER•I•A• P•AIG   50
VSI              -----RTS••••••VSAEE•RK•K•  •K• EAEIQ•VPQG•FV•A•YKFQ••RLY•••SNK••• A•AI•   40
ASI-I     (22)-LETNT•T•TS••••••VTAEQ•ETKIKE•MV D•QIQ•SPHD•FV•A•YNPK••RLY••ESNK•T• ••SIG   33
TI-I             LM•E•••M••••I•VPTKL•KE••GK•••SITNIPILLSG•PI•L•YL••••RLFDNILGF••Q M•V••   38
CI-1A     (17)-SIGASSA•TS•••V••MSAE••KE••L•DK•  •AQVE•IPVDAMVHLN•DPN••F•L•     A•A• ••T•G   35
CI-2A   <(14)-AGDRHNL•T•••••••KSVEE•KKV•LQDK• EAQI••LPVGTIV•MEY•I•••RLF••KLDNIAQ V•••G   33
LIE            TEFGSEL• SF••V••KTVDQ•REYFTLHY• QYDVYFLP•G•PV•L•L•YN••R•FYNPGTN••NHV•H•G   28
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Fig. 3. Amino-acid sequence of amaranth trypsin/subtilisin inhibitor ATSI. Comparison with inhibitors of the potato inhibitor I family: DI-1, trypsin inhibitor from fruits of wild tomato (*Lycopersicon peruvianum*) [3]; CMTI-V, Inhibitor of trypsin and activated Hageman factor from seeds of pumpkin (*Cucurbita maxima*) [7]; BGIA, Inhibitor of acidic amino-acid specific *Streptomyces griseus* proteinase from seeds of bitter gourd (*Momordica charantia*) [8]; VSI, subtilisin inhibitor from seeds of broad bean (*Vicia faba*) [5]; ASI-I, subtilisin inhibitor from seeds of adzuki beans (*Vigna angularis*) [6]; TI-I, Wound inducible leaf inhibitor from cultivated tomato (*Lycopersicon esculentum*) [4]; CI-1 and CI-2, chymotrypsin/subtilisin inhibitors from seeds of barley (*Hordeum vulgare*) [10,11]; LIE, proteinase inhibitor Eglin C from leech (*Hirudo medicinalis*) [12]. The numbering refers to ATSI. Note insertions in some sequences at position 30/31 and 64/65. Numbers in brackets indicate additional amino acids in the N-terminal of some inhibitors. (<) Denotes acetylated N-terminals and (— — —) indicates that the N-terminal of VSI has not been sequenced. (•) indicates identity with the sequence of ATSI. The % of amino-acid identity with ATSI is shown in the right column. P_1 in position 45 indicates the reactive site of the inhibitors.

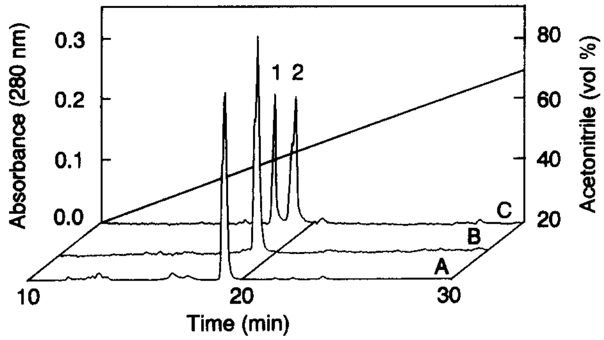

Fig. 4. Separation of ATSI peptides by RP-HPLC on a C_4 column. The column was equilibrated with 0.1% TFA and eluted with a linear gradient of acetonitrile in 0.1% TFA. (A) ≈ 200 μg ATSI + ATSIx (residues 1/3–69, pooled inhibitor from cation exchange (Fig. 1)); (B) ≈ 200 μg ATSI + ATSIx eluted from the subtilisin column at pH 3.2; (C) separation of B in C-terminal peptide (residues 46–69, peak 1) and N-terminal peptides (residues 1/3–45, peak 2) after reduction of the disulfide bond with 30 mM DTT in 6 M guanidine hydrochloride (pH 9.4) for 2 h at 25°C.

ments were detected (Fig. 4C). The peptide in the asymmetric peak 2 had an amino-acid composition and a mass spectrum corresponding to the expected mixture of N-terminal peptides of residues 1/3–45 (Table I). The material of peak 1 (≈ 0.4 nmol) was sequenced through 25 cycles with a repetitive yield of 90% (Fig. 3). The complete sequence of the C-terminal fragment was assigned starting with Asp^{46}-Phe-Arg and thus providing the necessary overlap with the N-terminal sequence. Amino-acid analyses and mass determinations confirmed that the entire C-terminal sequence was obtained (Table I). In the C-terminal fragment one-third of the residues are Val appearing twice in Val-Val-Val sequences (Fig. 3), which explains the relatively low recovery of this amino acid in the analyses (Table I).

Inhibitory activity

Inhibition of selected animal and microbial serine proteinases was first tested at a ATSI/enzyme ratio > 10 using specific *p*-nitroanilide substrates (Table II). Then the inhibitor constant K_i was determined for the significantly inhibited enzymes (Table II). Of the four pancreatic enzymes tested ATSI was a strong inhibitor of trypsin ($K_i \approx 0.34$ nM) and chymotrypsin ($K_i \approx 0.41$ nM), while elastase and kallikrein were not inactivated. ATSI was a very weak inhibitor of cathepsin G but leukocyte elastase was not affected. Among the nine

TABLE II

Inhibition of serine proteinases by ATSI

Inhibition was measured with a ATSI/enzyme molar ratio > 10. Pancreas and related leucocyte proteinases are in the top, proteinases related to blood clotting in the middle and microbial proteinases in the bottom of the table. K_m and the apparent inhibitor constant K_i' were determined for selected enzymes and K_i calculated. The table also contains information about enzyme source and substrates used.

Enzyme			Source [b]		*p*-NO_2-anilide substrate [c] (0.6 mM)	Residual enzyme activity [d] (%)	Inhibitor constant K_i (nm)
Name	Origin [a]	Conc. (nM)	Enz.	Subst.			
Chymotrypsin	h pancreas (α)	2.5	Si	KD	MS-Arg-Pro-Tyr-	4	0.41
Elastase	s pancreas	1	Bo	Si	S-Ala-Ala-Ala-	101	
Kallikrein	s pancreas	(0.1 U/ml)	Si	KD	D-Val-Leu-Arg-	87	
Trypsin	s pancreas	10	NN	KD	D-Val-Leu-Lys-	< 1	0.34
Cathepsin G	h leukocyte	5	NN	Si	S-Ala-Ala-Pro-Phe-	13	122
Elastase	h leukocyte	10	NN	Si	MS-Ala-Ala-Pro-Val-	97	
Factor VII_a	h plasma	300	NN	KD	D-Ile-Pro-Arg-	97	
Factor X_a	h plasma	3	WK	Ny	MCC-Gly-Arg-	96	
Factor XII_a	h plasma	30	IS	KD	D-Pro-Phe-Arg-	66	440
Kallikrein	h plasma	3	KD	KD	D-Pro-Phe-Arg-	90	
Plasmin	h plasma	10	KD	KD	D-Val-Leu-Lys-	10	38
Plasminogen activator	h urokinase	5	Se	KD	< Glu-Gly-Arg-	99	
Plasminogen activator	h tissue-type	5	NN	KD	D-Ile-Pro-Arg-	94	
Protein C_a	h plasma	5	NN	KD	< Glu-Pro-Arg	101	
Thrombin	h plasma (α)	(0.2 NIHU/ml)	NN	KD	D-Phe-Pip-Arg-	100	
Proteinase SP387	f	8	NN	Si	Bz-Arg-	48	
Proteinase SP226	b pumilus	0.5	NN	Ba	S-Ala-Ala-Pro-Phe-	83	
Subtilisin Carlsberg (A)	b licheniformis	2	NN	Ba	S-Ala-Ala-Pro-Phe-	28	5.5
Subtilisin Novo (BPN')	b amyloliquefaciens	4	NN	Ba	S-Ala-Ala-Pro-Phe-	5	0.37

[a] Source organism: b, bacillus; f, fusarium; h, human; s, swine.
[b] Supplier: Ba, Bachem; Bo, Boehringer; IS, gift from Dr. I. Schousboe, Copenhagen; KD, Kabi Diagnostics; NN, Novo Nordisk; Ny, Nycomed; Se, Serva; Si, Sigma; WK, gift from Dr. W. Kisiel, Albuquerque, NM.
[c] N-terminal blocking group: Bz, Benzoyl; < Glu, pyroglutamyl; MCC, methoxycarbonyl-D-cyclohexylanyl; MS, methoxysuccinyl; S, succinyl.
[d] All animal proteinases were tested with 800 μM ATSI, the microbial enzymes with 200 μM ATSI.

enzymes related to the the blood clotting system only plasmin was inhibited ($K_i \approx 38$ nM) and Factor XIIa (Hageman factor) weakly affected ($K_i \approx 440$ nM). Among the bacterial enzymes subtilisin NOVO (BPN′) was strongly inhibited ($K_i \approx 0.37$ nM), the effect on subtilisin Carlsberg was weaker ($K_i \approx 5.5$ nM) and the enzyme from *Bacillus pumilis* only slightly affected. A weak inhibition of a *Fusarium* enzyme (proteinase SP 387) with trypsin-like specificity was also observed (Table II).

The effect of various treatments on the inhibitory action of ATSI was tested. All subtilisin inhibitor activity was lost after incubation with porcine pepsin for 5 min at pH 2 and 25°C (14 Anson mU/mg inhibitor), while incubation for 1 h at pH 2 without enzyme did not result in inactivation. Heat treatments in the assay buffer for 10 min at temperatures up to 100°C did not affect the inhibitory activity of ATSI. As expected for an inhibitor following the canonical mechanism of inhibition [2,13] cleaved, unreduced ATSI (Fig. 4B) still possessed inhibitory activity towards the target enzymes, which was demonstrated by inhibitor staining with trypsin and subtilisin Carlsberg after isoelectric focusing (not shown).

Discussion

The high-purity ATSI obtained after cation exchange (Figs. 1 and 2) and RP-HPLC (Fig. 4) combined with the highly specific and, apparently, complete cleavage in the active site (Fig. 4) allowed for deduction of the complete 69 amino-acid sequence after two sequencer runs (Fig. 3). Analysis by mass spectrometry confirmed the result from amino-acid sequencing (Table I). The 28–51% similarity with previously sequenced inhibitors demonstrates that ATSI is a member of the potato inhibitor I family [2,13]. In accordance with the generally observed hypervariability in the P_1 position [2] members with specificity for trypsin (P_1 Lys), chymotrypsin (P_1 Leu, Met,Phe) and elastase (P_1 Ala, Leu) have been described (Fig. 3). As expected from the P_1 Lys ATSI is a strong inhibitor of porcine trypsin ($K_1 \approx 0.34$ nM) and, apparently, more efficient than the two previously identified members of the family with Lys as the P_1 amino acid [3,7], DI-1 from fruits of wild tomato ($K_i \approx 7.2$ nM) and CMTI-V from pumpkin seeds ($K_i \approx 16$ nM). DI-1 was not inhibitory towards chymotrypsin. CMTI-V was also an inhibitor of Factor XIIa (Hageman factor, $K_i \approx 41$ nM), but not of kallikrein. No other enzymes were tested as target proteinases for these inhibitors. ATSI was found to be a weak inhibitor of Factor XIIa and of plasmin, but several other enzymes with specificity for basic amino acids in the S_1 position were not affected (Table II). In contrast to DI-1, ATSI is also an inhibitor of enzymes with specificity for aromatic amino acids, including chymotrypsin ($K_i \approx 0.41$ nM) and cathepsin G, while elastases were not affected. Previously, a few strong trypsin inhibitors from other families with P_1 Lys or Arg have been found to inhibit chymotrypsin at the same reactive site [2] confirming that interactions apart from those in the enzyme S_1 subsite may contribute significantly to the enzyme–inhibitor binding affinity. ATSI is a potent inhibitor of microbial proteinases, including subtilisins and *Fusarium* proteinase (Table II), as well as *Aspergillus* proteinase and proteinase K (isoelectric focusing, not shown). All previously tested members of the inhibitor I family with P_1 Leu, Met or Ala are also inactivating microbial alkaline proteinases. In comparison with the trypsin subfamily, these enzymes have an extended binding cleft with at least eight subsites which may interact with individual amino acids of the inhibitor loop thus making the P_1 contact even less important [13]. Our results support the generally accepted assumption that the inhibitors of the family are parts of antimicrobial defence systems [1].

The tendency to form defined aggregates (Fig. 2B) has been observed previously in the inhibitor I family, e.g. with the potato and barley inhibitors, and so has molecular heterogeneity due to formation of molecules (like ATSIx) with 'ragged' amino terminals [5,11]. Like several members of the family (Fig. 3) ATSI contains a disulphide bridge connecting the N-terminal Cys-4 with Cys-49 in the reactive-site loop hinge region. However, many of the inhibitors lack Cys in their sequence (Fig. 3) and loop stabilization as well as the high stability towards heat and denaturing agents seem to be based solely on non-covalent interactions in the inhibitor I family [13]. With respect to highly conserved regions of the sequence (Fig. 3) ATSI is very similar to the other members of the family and, undoubtedly, the general conformation is as found for barley CI-2 and leech Eglin [2,13]. Studies of complexes between these two inhibitors and cognate enzymes (chymotrypsin and subtilisin) have demonstrated that a prerequisite for proper inhibition is the establishment of a network of hydrogen bonds between the loop residues and sidechains extending from the protein core [13]. Especially important are the two Arg residues in the highly conserved Arg^{51}-Val-Arg^{53} sequence of the previously sequenced inhibitors (Fig. 3). The interaction between Arg-53 and the P_2 residue (Thr) has directly been suggested to parallel the loop-stabilizing disulfide bond between P_2 (or P_3) and the inhibitor core in other inhibitor families [14]. Experiments with engineered forms of Eglin [14] and CI-2 [15] where Arg-53 was substituted by Lys or Ala, respectively, resulted in decreased inhibitor potential and/or protein destabilization due to loss of essential hydrogen bonds. In ATSI amino acid 53 is Trp, a bulky hydrophobic residue with capacity for hydrogen bond formation, apparently, without adverse effects on inhibitor capacity or heat stability. The only other known natural inhibitor without Arg in this position is CI-1 with Phe-53. This atypical member of the family, which has P_2 Pro or His and also lacks residues 57–60 (Fig. 3), is a much weaker chymotrypsin inhibitor than CI-2 using casein as substrate and is readily inactivated at ≈ 70°C [9]. A common property of ATSI (Table II) and CI-1 [22] is the very weak effect on the *Bacillus pumilus* enzyme which was strongly inhibited by both barley CI-2 [22] and faba bean VSI [5].

In conclusion, our results suggest that structural characterization of ATSI or Trp-53 engineered mutants of the well-characterized inhibitors eglin or CI-2 might further elucidate the non-covalent loop stabilization in the potato inhibitor I family. Although ATSI is very heat stable it was immediately inactivated by pepsin at pH 2 and the inhibitor hardly represents an antinutritional factor in foods or in feeds for monogastric animals.

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